



Text-only synthesis for zero-shot visual captioning

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ABSTRACT

From paired image-text training to text-only training for visual captioning, the pursuit of relaxing the requirements for high-cost and large-scale annotation of good quality data remains consistent. In this paper, we propose Text-only Synthesis for Visual Captioning (ToCa), which further advances this relaxation with less human labor and lower computational cost. Specifically, we deconstruct caption text into structures and lexical words, which serve as the fundamental components of the caption. By combining different structures and lexical words as inputs to the large language model, massive captions that contain various patterns of lexical words are generated. This method not only approaches the target domain but also surpasses it by generating new captions, thereby enhancing the zero-shot generalization ability of the model. Considering the different levels of data access in the real world, we define three synthesis scenarios: cross-domain synthesis, in-domain synthesis, and data-efficient synthesis. Experiments in these scenarios demonstrate the generalizability, transferability and practicability of ToCa with a nearly 5 CIDEr improvement for zero-shot cross-domain captioning (without any access to the target-domain training captions) and a maximum increase of over 20 CIDEr for data-efficient captioning. The code and data will be available at <https://github.com/mrazhou/ToCa>.

1. Introduction

Visual captioning [1–3] aims to describe the entities in an image and their interactions using natural language, i.e., what the entities are and how they interact with each other. Based on extensively annotated paired data, deep learning models [4–8] have demonstrated remarkable performance, fostering applications in the fields of image retrieval and visually impaired assistance [9], among others. However, the annotation of paired image-text data on a large scale presents significant challenges in terms of cost, labor, and time, impeding the widespread application and generalization of models across various scenarios.

To address these challenges, on the one hand, text-only training methods [10–14] leverage the shared semantic representation of CLIP [15] to train zero-shot captioning models using only text data, significantly reducing the difficulty of constructing training data. On the other hand, the paired image-text synthesis methods [16,17] utilize diffusion models [18] and large language models (LLMs) [19] to generate a large volume of image-text data for training at a remarkably low human cost. However, challenges persist, and the former still requires extensive task-specific text data through manual efforts, while the latter necessitates substantial computational resources to simultaneously generate image-text pairs.

In this paper, we introduce the text-only synthesis for captioning which aims to further mitigate the need for manually curated datasets and advance the field from paired image-text synthesis to text-only synthesis. Synthesizing text is simple, but crafting desired text requires skills. In order to generate desired text in a controlled manner, we propose Text-only Synthesis for Visual Captioning (ToCa). Firstly, the captions are deconstructed into meaningful lexical word pairs (representing entities) and structure templates (representing their interactions). Subsequently, LLM is used to generate captions that encompass various interactive behaviors involving these entities. By approaching the target domain, ToCa simultaneously produces novel captions that enhance the model's ability to generalize beyond its original target domain.

ToCa exhibits several distinctive features that contribute to its utility and performance in text synthesis: (1) *Data efficiency*, ToCa can require only a small corpus (even 56 captions) to synthesize a large volume of text data, significantly enhancing model generalization capabilities. (2) *Open accessibility*, the LLM used is open-source and can operate on most graphics cards. This accessibility not only safeguards data privacy but also supports broader usage. Both the code and data will also be openly accessible. (3) *Flexibility*, ToCa allows for the flexible synthesis of any desired text data using different lexical word pairs

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and structure templates. (4) *Reduced computational steps*, compared to paired image–text synthesis methods, ToCa eliminates the step of image generation (a time-consuming diffusion model generation process), requiring only a 7B LLM to produce text data that meets requirements.

To validate and showcase the formidable synthesis capabilities of ToCa, we conduct a series of extensive experiments based on the accessibility of the data in the real-world. These experiments are categorized as follows: In-domain synthesis, both the data from the corpus and the target dataset are accessible; Cross-domain synthesis, only the data from the corpus is accessible, while the target dataset remained inaccessible; Data-efficient synthesis, limited access to the corpus and the target dataset.

Our main contributions are the following:

- We propose the concept of text-only synthesis for captioning, aiming to explore methods that require fewer human resources and less computational steps to generate training data for captioning models.
- We present ToCa, an algorithm for text synthesis that is flexible, efficient, and completely open-source. It is designed based on the linguistic composition of caption and greatly improves the model’s zero-shot generalization performance.
- We establish three common application challenges in real-world, and through these challenges, we validate and demonstrate the impressive synthesis capabilities of ToCa. ToCa explores a purely text-only synthetic data generation paradigm for visual captioning: unlike paired image–text synthesis approaches, it requires no image-generation step, relying solely on lexical-pair and structure-template recombination with an LLM.

2. Related works

2.1. Paired image–text training

Traditional captioning methods [20–23] typically require training on image–text aligned data. During this period of research, the prevalent approach involves employing an encoder–decoder architecture and making subsequent improvements upon it. This involves a transition from CNN-based [24] visual encoders to ViT-based [25] ones, as well as a shift from LSTM-based [26] language decoders to Transformers. Various mechanisms [27,28] are explored to enhance the alignment of visual and linguistic features. However, the application of these models is limited by the high cost and scalability challenges associated with manual annotation of paired image–text data. Furthermore, there is a lack of investigation into the zero-shot generalization capabilities of these methods.

2.2. Text-only training

To address the challenge of collecting paired image–text data and the limited generalization of zero-shot visual captioning, text-only training has garnered extensive attention [29]. CLIP-guided text-only training has evolved from early paradigms, such as ZeroCap [30], MAGIC [10], CapDec [11], and DeCap [12], toward more refined modality alignment strategies. Specifically, DeCap projects visual features into the CLIP text embedding space and incorporates an extra-training support memory to mitigate the modality gap. MacCap [31] further analyzes the CLIP latent space, employing sub-region feature aggregation, noise injection, and re-ranking techniques to significantly enhance both zero-shot captioning performance. MeaCap [32] integrates a “training-free + text-only training” strategy, utilizing text memory and a retrieval-filtering mechanism to alleviate hallucinations while preserving general knowledge. Collectively, these works demonstrate that leveraging pre-trained VLMs or LLMs with solely text corpora can yield competitive zero-shot results on benchmarks. However, these

approaches typically rely on existing substantial amounts of manually annotated text data, and their generalization is often constrained by the limitations inherent to human-generated data. In this paper, we propose a method for synthesizing text data, enabling AI-generated content (AIGC) for text-only training to further enhance zero-shot generalization abilities.

2.3. Paired image–text creation

Another solution to address the high cost of paired image–text annotation is to leverage automated methods to create image–text data. This approach generally falls into two categories: collecting existing data through web scraping [33,34] and synthesizing data using generative models [16,17]. Regarding synthesis, SynTIC [16] and PCM-Net [35] leverage T2I priors (such as Stable Diffusion) to synthesize images. By employing techniques like feature Mix-up and CLIP-weighted loss to mitigate the semantic mismatch between synthetic images and text, these approaches have achieved advanced zero-shot captioning. Similarly, ICSD [17] utilized Stable Diffusion and the ChatGPT3.5 API to synthesize paired data. However, these approaches face significant challenges. For web scraping, recent research has specifically investigated the copyright risks and privacy audits associated with T2I training data [36]. Studies proposing black-box audit frameworks and empirical analyses highlight the legal and compliance uncertainties inherent in large-scale scraped data. For synthesis, relying on commercial products like ChatGPT3.5 (as in ICSD) raises data privacy issues and potential commercial risks due to their closed-source nature. Furthermore, the generation process introduces additional computational costs, where the effectiveness of the Stable Diffusion and LLM components directly affects both the quality and time efficiency of the data synthesis.

2.4. Large language models

Following the impressive demonstration of language proficiency by OpenAI’s ChatGPT, the open-sourcing of LLM has garnered significant attention and research [37,38]. Apart from the pioneering open-source model Llama [39], Mistral has also released their model with high performance and efficient inference [40]. The technique of model quantization [41,42] has further enhanced the usability of LLM. These developments have led us to choose the quantified Mistral and Llama-2 as the LLM for our research in this paper, aiming to further enhance the usability of our methods.

3. Method

3.1. Preliminaries

Unlike supervised methods, text-only training assumes that only the text is accessible during training. Given the powerful image–text alignment representation capability and zero-shot ability of CLIP, for a given text T and image I , the corresponding embeddings E_T and E_I can be obtained in the same representation space via the CLIP encoder. During training, the accessible text is encoded by CLIP to obtain E_T , which is then decoded back into the original text T using a text decoder composed of a lightweight mapping network and a pre-trained autoregressive language model. In essence, text-only training is a process of training a text autoencoder, or more precisely, a reconstruction of the input text from CLIP’s textual embedding. Since the training objective is self-reconstruction (the input and output are both T), no labels are required, significantly reducing data costs. During the inference phase, the input image is encoded by the CLIP encoder to obtain the image embedding E_I , which is then used to generate the caption.

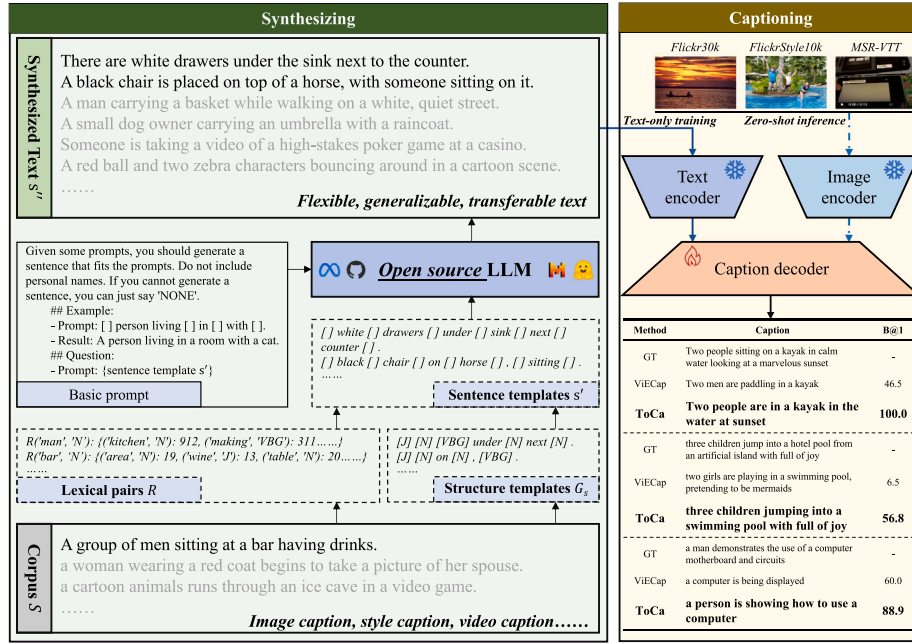


Fig. 1. Visualization of the process and results of synthesizing and captioning. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

3.2. Problem definition

For text-only synthesis, given the corpus S , domain prior knowledge $\mathcal{X}$, where $\mathcal{X}$ could be entities or caption datasets, and the target domain dataset $\mathcal{T}$, our objective is to synthesize a dataset $\mathcal{D}$ that closely approximates or even encompasses the distribution of $\mathcal{T}$ by constructing a synthetic generation strategy $\mathcal{F}$. Formalized as follows:

$$\min(\mathcal{L}(\mathcal{D}, \mathcal{T}) - \mathcal{L}(\mathcal{D} \setminus \mathcal{T}, \mathcal{T})), \quad (1)$$

s.t. $\mathcal{D} = \mathcal{F}(S, \mathcal{X})$,

where $\mathcal{L}$ represents a measure of the distance between the centers of two data distributions. The term $\min(\mathcal{L}(\mathcal{D}, \mathcal{T}))$ ensures that the generated data $\mathcal{D}$ is close to $\mathcal{T}$, thereby guaranteeing domain-specific performance. Additionally, the term $\min - \mathcal{L}(\mathcal{D} \setminus \mathcal{T}, \mathcal{T})$ aims to incorporate information in the generated data $\mathcal{D}$ that is not present in $\mathcal{T}$, thereby enhancing domain-agnostic generalization. Finding a suitable $\mathcal{L}$ is indeed a challenging task, but it is not the focus of this research. Instead, we evaluate the synthesis performance by examining the statistical measures between $\mathcal{D}$ and $\mathcal{T}$ (see Section 4.7), as well as the caption performance of the final model. These evaluations provide insights into the effectiveness of the synthesized data.

Taking into account the challenges arising from variations in data accessibility in real-world, we classify text synthesis into three distinct types based on the availability of S and $\mathcal{X}$, as illustrated conceptually in Fig. 2.

In-domain synthesis: in this scenario, we have access to both S and the target domain data, i.e., $\mathcal{X} = \mathcal{T}$. The objective is to synthesize $\mathcal{D}$ with a focus on generating novel data that approaches and even surpasses $\mathcal{T}$.

Cross-domain synthesis: in this case, the target dataset $\mathcal{T}$ is entirely inaccessible; only a corpus S from a different, accessible domain (e.g., Flickr30k when the target is COCO) can be used. We further distinguish two sub-cases: (i) a strict, zero-target-information variant with $\mathcal{X} = \emptyset$, in which no information whatsoever about $\mathcal{T}$ is used, and which we treat as the primary evidence of cross-domain transferability (Section 4); and (ii) an optional weak-prior variant $\mathcal{X} = \mathcal{R}$, in which a small, sparse table of lexical-pair co-occurrence statistics extracted from $\mathcal{T}$'s captions is additionally available. Importantly, $\mathcal{X} = \mathcal{R}$ carries

substantially less information than $\mathcal{T}$ itself: it only records which content words tend to co-occur (a token-pair frequency table), without any sentence-level structure, word order, or stylistic information that would make it directly usable as training data (quantified in Section 4). We report this weak-prior variant only as an oracle/sensitivity study on how much additional lexical co-occurrence information can help when such statistics happen to be available; since $\mathcal{X} = \mathcal{R}$ is itself extracted from $\mathcal{T}$'s captions, this variant necessarily assumes some access to target-domain text and is not part of the strict, data-inaccessible cross-domain claim, which relies solely on ToCa_S.

Data-efficient synthesis: in this situation, both S and $\mathcal{X}$ are simultaneously limited to a very small subset of the target domain (i.e., $S = \mathcal{X} \subset \mathcal{T}$, potentially as few as several dozen sentences). The objective is to synthesize a much larger $\mathcal{D}$ ($|\mathcal{D}| \gg |S|$) via combinatorial recombination of the limited lexical pairs and structure templates, thereby efficiently amplifying a tiny seed corpus.

3.3. Text synthesis

Notation. For clarity, Table 1 summarizes the main symbols used throughout this subsection.

Lexical words carry the majority of the semantic information in a sentence, while structure helps to express the relationships between lexical word and construct coherent sentences. In the context of captions, the essence lies in describing the entities and their interactions within an image. Therefore, our core insight is to deconstruct a caption text into two fundamental linguistic elements: lexical pairs and structure templates. By recombining these elements and inputting them into a knowledge-rich LLM, we can achieve the expansion of in-domain text data and the creation of out-of-domain text. Specifically, the text synthesis process as shown in Fig. 1 involves three steps: structure template construction, lexical pair extraction, and LLM text synthesis, and the overall procedure is summarized in Algorithm 1. **Structure template construction.** Here we define that a structure template G_s refers to the basic structure of a sentence s , including most function words and part-of-speech (POS) of lexical words, such as "[N] [VBG] [N] on [N]". For a word *token*, we can write the template function

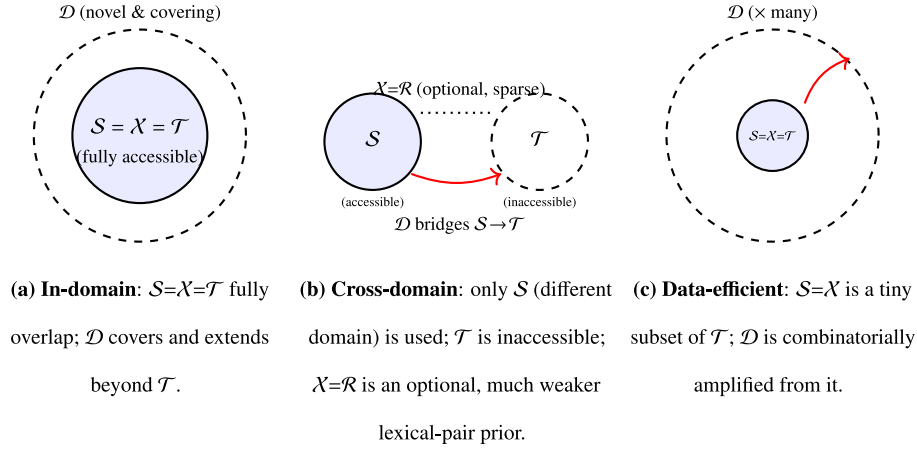


Fig. 2. Conceptual illustration of the three data-accessibility scenarios studied in this paper. Solid regions denote data that is accessible during synthesis; dashed regions denote data that is inaccessible. Formal definitions are given below.

Table 1

Notation used in Section 3.

Symbol	Meaning
$S, \mathcal{X}, \mathcal{T}, \mathcal{D}$	Accessible corpus, prior knowledge, target set, synthesized set
$F, \mathcal{L}$	Synthesis strategy; distance measure between distributions (Eq. (1))
$\mathcal{Q}_c, \mathcal{Q}_f$	POS tag sets for content (lexical) words and function words
$g(\cdot)$	Template function mapping a token to a function word or a POS placeholder
$G_s, \mathcal{G}$	Structure template of sentence s ; corpus-level template distribution
N_{G_s}	Occurrence count of template G_s in S
$R_s(\cdot), \mathcal{R}$	Sentence-level and corpus-level lexical-pair co-occurrence statistics
$N_{token}, N_{token_s, token_j}$	Occurrence count of a lexical token/a lexical pair in S
w_i	Shorthand for the i th sampled lexical word $token_{s, g_i}^i$ (used in Alg. 1)
τ	Diversity (temperature) hyperparameter, Eq. (12)
s, s', s''	original sentence; masked sentence template; LLM-completed sentence
m	preset target size/upper bound of the synthesized dataset $\mathcal{D}$ (Eq. (13))

accordingly:

$$g(token) = \begin{cases} token & \text{if } q_{token} \in \mathcal{Q}_f, \\ [q_{token}] & \text{if } q_{token} \in \mathcal{Q}_c, \\ None & \text{if else,} \end{cases} \quad (2)$$

where q_{token} is the POS of the token, $\mathcal{Q}_c$ is the set of lexical words, and $\mathcal{Q}_f$ is the set of function words. For a sentence s , we can obtain its structure template G_s :

$$G_s = [g(token_s^0), g(token_s^1), \dots, g(token_s^n)], \quad (3)$$

where n is the token length of sentence s .

We employ the use of NLTK² to identify the POSs of words. By conducting a statistical analysis of the structure templates of all sentences in the corpus S , we can obtain distributional information about these templates:

$$\mathcal{G} = \{(G_s, N_{G_s}) | s \in S\}, \quad (4)$$

where N_{G_s} is the number of occurrences of structure template G_s in S . **Lexical pair extraction.** The set $R(token^i)$ represents the collection of lexical relationships concerning a specific lexical item $token^i$. It signifies the association between $token^i$ and the set of lexical items that appear

after it in the same sentence. By gathering lexical relationships from the corpus, we can more comprehensively restore the entity information and relationships between entities within the sentence. Specifically, given a sentence s in the corpus S , we can obtain its set of lexical pairs:

$$R_s(token_s^i) = \{(token_s^j, N_{token_s^i, token_s^j}) | token_s^i, token_s^j \in s, q_{token_s^i}, q_{token_s^j} \in \mathcal{Q}_c, i < j\} \quad (5)$$

where $N_{token_s^i, token_s^j}$ is the number of occurrences of the lexical pair $(token_s^i, token_s^j)$ in the sentence s . Similarly, by statistically analyzing the lexical pairs of all sentences, we can obtain the distribution information of lexical pairs $\mathcal{R}$:

$$R(token, q_{token}) = \{R_s(token) | s \in S, token \in s, q_{token} \in \mathcal{Q}_c\}, \quad (6)$$

$$\mathcal{R} = \{(R(token, q_{token}), N_{token}) | s \in S, token \in s, q_{token} \in \mathcal{Q}_c\}, \quad (7)$$

where N_{token} is the number of occurrences of the lexical word $token$ in the corpus S .

LLM text synthesis. Synthesizing text using LLMs is straightforward, yet producing text on a massive scale that meets specific requirements demands sophisticated algorithmic strategies. On one hand, generating a substantial amount of data necessitates a substantial number of prompts. However, we do not wish to manually create a vast number of prompts, as it goes against our original intention. On the other hand, different types and sizes of LLMs possess varying levels of understanding and adherence to prompts. Inadequate prompts result in low-quality desired text. The key lies in having a sufficient quantity of prompts that LLMs can easily follow. By utilizing the structure templates and lexical pairs constructed in the previous context, the ToCa can swiftly generate large-scale prompts through recombination. Another crucial aspect of ToCa is guiding the LLM to complete sentences based on sentence templates, rather than merely constructing them.

Specifically, the process begins by randomly sampling $G_s = [g_s^0, g_s^1, \dots, g_s^n]$ from the structure template $\mathcal{G}$, based on the size of N_{G_s} . Next, for the POS of the first lexical word g_s^0 in G , a random sampling is performed on the lexical words according to the size of N_{token} to obtain w_0 (shorthand for $token_{s, g_s^0}^0$; see Table 1) with the category g_s^0 .

Subsequently, g_s^0 in G_s is replaced with w_0 . Following this, the lexical category g_s^1 of w_0 is sampled based on $R(w_0, g_s^1)$, and then substituted for g_s^1 in G_s . This entire process can be represented using the following formula:

$$w_0 \sim \frac{N_{w_0}}{\sum_{w_0} N_{w_0}}, \quad (8)$$

² <https://www.nltk.org/>.

$$w_1 \sim \frac{N_{w_0, w_1}}{\sum_{w_1} N_{w_0, w_1}}. \quad (9)$$

For the third and subsequent lexical categories g_f^i , it is obvious that their probability distribution should be related to all previous tokens. We define $[token^{i_0}, token^{i_1}, \dots, token^{i_k}]$ to represent these lexical tokens appearing in the sentence in the order of $i_0 < i_1 < \dots < i_k$. Then, writing w_i for the i th sampled lexical word (i.e. $token_{s, g_f^i}^i$):

$$w_i \sim P(w_i | [w_0, \dots, w_{i-1}]). \quad (10)$$

When i is large, accurately computing the conditional probability in Eq. (10) becomes arduous and inefficient. Hence, we employ a straightforward yet potent approximation assumption: given that w_i is established, the probabilities of prior token occurrences are independent of their sequence and the events of these tokens occurring are mutually independent. Thus, we can approximate Eq. (10) as follows:

$$\begin{aligned} & \frac{P(w_i | [w_0, w_1, \dots, w_{i-1}])}{P([w_0, \dots, w_{i-1}] | w_i) P(w_i)} \\ &= \frac{P([w_0, \dots, w_{i-1}])}{P(w_i) \prod_{j=0}^{i-1} P(w_j | w_i)} \\ &= \frac{P(w_0 \cap w_1 \cap \dots \cap w_{i-1})}{(\prod_{j=0}^{i-1} N_{w_j, w_i}) / N_{w_i}^{i-1}} \\ &= \frac{\sum_{w_i} (\prod_{j=0}^{i-1} N_{w_j, w_i}) / N_{w_i}^{i-1}}{\sum_{w_i} (\prod_{j=0}^{i-1} N_{w_j, w_i}) / N_{w_i}^{i-1}}. \end{aligned} \quad (11)$$

From Eq. (11), it can be seen that $N_{w_i}^{i-1}$ suppresses the excessive sampling of high-frequency words, enhancing the diversity of the data. To finely adjust the diversity of the data, we introduce a hyperparameter τ to control the degree of influence of $N_{w_i}^{i-1}$, where a larger τ leads to less diversity in sentences:

$$w_i \sim \frac{(\prod_{j=0}^{i-1} N_{w_j, w_i}) / N_{w_i}^{(i-1)/\tau}}{\sum_{w_i} (\prod_{j=0}^{i-1} N_{w_j, w_i}) / N_{w_i}^{(i-1)/\tau}} \quad (12)$$

Through the above derivation, we can replace each lexical token in G_s according to the approximate probability distribution of Eq. (12), thereby obtaining a sentence s containing many lexical tokens. To utilize the generation ability of LLM, we add MASK markers on both sides of all real words in the obtained s , thus obtaining the final sentence template s' , such as "[] horse [] sitting [] field [] on [] background []". Then, we can input s' into LLM, requiring LLM to complete the gaps.

Now, from the output of the LLM, we can obtain a complete sentence s'' that closely aligns in both style and content with our desired sentence. If s'' does not contain all $\{w_0, \dots, w_n\}$, it is likely that s'' contains some conflicting information. In this case, we can simply filter out s'' , which is equivalent to indirectly utilizing the filtering ability of LLM. By repeating the above process, we can obtain the synthetic dataset $D = s''_0, s''_1, \dots, s''_m$, where m is the preset size of the synthetic dataset. The upper bound of m is given by:

$$m \leq \sum_{i=1}^{|G|} \prod_{j=1}^{n_i} k_{i,j}, \quad (13)$$

where n_i represents the number of lexical items in the i th template. $k_{i,j}$ represents the size of optional lexical items for the j th $token \in Q_c$ in the i th template.

Algorithm 1 summarizes the complete text-synthesis procedure described above, with explicit correspondence to the equations introduced in this subsection.

4. Experiment

ToCa generates text data in three scenarios: in-domain synthesis, cross-domain synthesis, and data-efficient synthesis. Then assess the

Algorithm 1 ToCa Text Synthesis

Require: Corpus S , POS sets Q_c, Q_f , diversity hyperparameter τ , target size m

Ensure: Synthesized dataset D

- 1: Extract structure templates $\mathcal{G} = \{(G_s, N_{G_s})\}$ from S {Eqs. (2)–(4)}
- 2: Extract lexical-pair statistics $\mathcal{R}$ from S {Eqs. (5)–(7)}
- 3: $D \leftarrow \emptyset$
- 4: **while** $|D| < m$ **do**
- 5: Sample $G_s \sim \mathcal{G}$ weighted by N_{G_s}
- 6: Sample first lexical word w_0 weighted by N_{w_0}
- 7: **for** each remaining lexical slot $i = 1, \dots, n$ **do**
- 8: Sample w_i from approximate conditional distribution in Eq. (12) (temperature τ)
- 9: **end for**
- 10: Insert mask tokens around $\{w_0, \dots, w_n\}$ in G_s to obtain sentence template s'
- 11: Query the LLM to complete s' , obtaining candidate sentence s''
- 12: **if** s'' contains all of $\{w_0, \dots, w_n\}$ **then**
- 13: $D \leftarrow D \cup \{s''\}$ {filtering step, Sec. 4 reports the filtering rate}
- 14: **end if**
- 15: **end while**
- 16: **return** D

performance of models trained on synthesized data in zero-shot visual captioning tasks. This evaluation demonstrates the algorithm's generalizability, transferability, and practicability in various contexts.

Implementation details. We employ the text-only training method ViCap [14] as the foundational caption model, keeping the training configuration aligned with it. The model undergoes an initial training of 15 epochs on the synthetic data, followed by fine-tuning on the accessible set $\mathcal{X} = \mathcal{T}$ (i.e. $D + \mathcal{T}$) and without any fine-tuning on $\mathcal{X} = \mathcal{R}$ (i.e. D). Regarding the LLM used for synthetic data generation, the 8-bit quantized Mistral-7B on huggingface³ is used by default. It demands approximately 8 GB of GPU memory. When executed on an NVIDIA GeForce RTX 3090 GPU, it takes approximately 1 s to generate a single sentence. Leveraging the advantages of quantization technology, a 3090 GPU (24 GB memory) enables the simultaneous execution of two processes, thereby accelerating the generation process. For text generation via the LLM, we use the following sampling parameters: a low temperature of $\tau_{llm} = 0.2$, top- $p = 0.95$, top- $k = 40$, and a repetition penalty of 1.1. During the captioning evaluation phase, we adopt standard beam search with a beam size of 5 and a length penalty of 1.0 across all benchmarks to ensure consistency.

Datasets and metrics. Experiments are conducted using the widely recognized visual caption benchmarks, COCO [43], NoCaps [44], and Flickr30k [45], to assess the performance enhancement brought about by the generated data in in-domain captioning and cross-domain captioning tasks. By default, we utilize COCO as the corpus. COCO test set is used for in-domain caption evaluation. Flickr30k and NoCaps Val are used for cross-domain caption evaluation. The dataset partitioning remains consistent with previous works [11,14] to ensure fairness. The metrics employed are the commonly used BLEU@n (B@n) [46], METEOR (M) [47], CIDEr (C) [48], and SPICE (S) [49].

4.1. In-domain synthesis

In this section, we simulate and process real-world scenarios where established or historical data are already created, i.e. $S = \mathcal{X} = \mathcal{T} = \text{COCO training set}$. In such cases, ToCa performs in-domain synthesis, with the objective of generating novel caption texts to enhance the model's in-domain captioning performance and cross-domain caption generalization performance.

³ <https://huggingface.co/TheBloke/Mistral-7B-Instruct-v0.2-GGUF>.

Table 2
Generalizable in-domain synthesis.

Methods	Cross-domain captioning								In-domain captioning			
	Flickr30k				NoCaps Val (CIDEr)				COCO			
	B@4	M	C	S	In	Near	Out	Overall	B@4	M	C	S
<i>Paired image-text training, zero-shot inference</i>												
I-Tuning _{Base} [22]	–	–	–	–	83.9	70.3	48.1	67.8	34.8	28.3	116.7	21.8
SmallCap ^a [23]	–	–	–	–	83.3	77.1	65.0	75.8	37.0	27.9	119.7	21.3
<i>Paired image-text synthesis, zero-shot inference</i>												
ICSD [17]	–	–	–	–	–	–	–	–	29.9	25.4	96.6	–
SynTIC [16]	17.9	18.6	38.4	11.9	–	–	–	–	29.9	25.8	101.1	19.3
<i>Text-only training, zero-shot inference</i>												
DeCap ^b [12]	16.3	17.9	35.7	11.1	65.2	47.8	25.8	45.9	24.7	25.0	91.2	18.7
MacCap ^b [31]	15.4	17.9	30.3	11.4	–	–	–	–	17.4	22.3	69.7	15.7
SRACap [29]	18.6	18.0	43.2	11.7	61.5	64.5	64.0	65.8	26.2	24.2	91.5	17.7
CapDec [11]	17.3	18.6	35.7	–	60.1	50.2	28.7	45.9	26.4	25.1	91.8	–
+ToCa	<u>17.9</u>	<u>18.9</u>	40.4	<u>12.6</u>	<u>70.0</u>	60.3	44.3	57.7	<u>27.1</u>	<u>25.8</u>	<u>94.8</u>	<u>19.3</u>
ViECap [14]	17.4	18.0	38.4	11.2	61.1	64.3	65.0	66.2	<u>27.2</u>	24.8	92.9	18.2
+ToCa _{llama2-7b}	17.7	<u>18.7</u>	<u>42.7</u>	<u>12.3</u>	62.5	<u>67.8</u>	<u>69.1</u>	<u>69.5</u>	<u>27.2</u>	25.3	94.0	18.8
+ToCa	<u>18.2</u>	<u>18.7</u>	<u>43.9</u>	<u>12.6</u>	<u>64.6</u>	<u>69.1</u>	<u>70.5</u>	<u>70.9</u>	<u>27.1</u>	<u>25.4</u>	<u>95.0</u>	<u>19.0</u>

^a Memory bank.

^b Frozen language model.

Generalizability. As depicted in Table 2, the ViECap, trained with augmentation through ToCa, exhibits remarkable superiority in both in-domain captioning and cross-domain captioning when compared to other text-only training methods. Notably, it is worth highlighting that ToCa provides ViECap with a gain of nearly 5 points in the CIDEr metric for cross-domain captioning. Additionally, in the NoCaps out-of-domain setting, ToCa achieves a significant advancement over the paired image-text training method, surpassing SmallCap by 5.5 points. Furthermore, it is noteworthy that even though we employ the COCO training set for the extraction of lexical pairs and the construction of structural templates to synthesize text, there is an improvement of 2.1 points in the CIDEr score on the COCO test set. These enhancements demonstrate the effectiveness of ToCa in utilizing the information within the corpus to generate novel, diverse text, thereby infusing the model with out-of-domain knowledge and enhancing its generalization capabilities. Even compared with the paired image-text synthesis method SynTIC, ToCa has stronger cross-domain generalization. We also compare against MacCap [31], a more recent zero-shot text-only method, in Table 2: ToCa (+ViECap) substantially outperforms it under the same in-domain protocol (COCO CIDEr 95.0 vs. 69.7), indicating that ToCa’s advantage holds even against this stronger, more recent baseline. The small performance gap between ToCa_{llama2-7b} and ToCa also indicates that the proposed method has a certain degree of universality to different LLMs.

Flexibility. To demonstrate the flexibility and controllability of ToCa in text synthesis, we apply it to the tasks of style captioning (FlickrStyle10k [50]) and video captioning (MSR-VTT [51]). From Table 3, it is evident that ToCa significantly improves the model’s performance in both romantic and humorous captioning. This underscores the flexibility and controllability of ToCa in style captioning. From Table 3, it can be observed that ToCa consistently enhances captioning performance even in the context of video captioning tasks. This indicates the versatility of ToCa across different tasks. We attribute these improvements to the construction of task-independent lexical pairs and structural templates by ToCa. By deconstructing captions from a linguistic perspective, ToCa enables flexible synthesis of any desired text.

4.2. Cross-domain synthesis

In this section, we explore the impact of synthesized text on cross-domain captioning when the target dataset $\mathcal{T}$, specifically the COCO training set, is entirely inaccessible: no COCO images, no COCO captions, and no statistics derived from COCO captions are used at any stage of the pipeline. Only a certain amount of similar corpus S ,

such as Flickr30k, is accessible, and ToCa_S extracts all of its structure templates $\mathcal{G}$ and lexical pairs $\mathcal{R}$ solely from S . As shown in Table 4, this strictly zero-target-information ToCa_S continues to significantly outperform other models, exhibiting a CIDEr score advantage of 4.7 over ViECap. This result is the primary evidence of the remarkable cross-domain **transferability** of ToCa, and is the basis for the cross-domain improvement reported in the abstract and introduction.

Effect of an additional weak lexical prior. We additionally report a separate sensitivity study, ToCa_{S+ $\mathcal{X}$} , in which a small, sparse prior $\mathcal{X} = \mathcal{R} - \mathcal{R}$ – a lexical-pair co-occurrence table extracted from the COCO training captions – is made available in addition to S . We stress that $\mathcal{X} = \mathcal{R}$ is a qualitatively much weaker signal than the full target corpus $\mathcal{T}$: each entry is merely an unordered co-occurrence count between two content words, carrying no word order, syntactic structure, or narrative style, and consequently cannot be used to reconstruct anything resembling an actual caption sentence – unlike $\mathcal{T}$, whose sentences could be used directly as training data. Concretely, the lexical-pair table extracted from the full COCO training set contains 1,202,172 distinct token-pair entries, versus 566,747 full sentences in $\mathcal{T}$ (i.e., the pair table has more, but categorically less informative, entries than $\mathcal{T}$ has sentences, since each of the 566,747 sentences contributes several pairwise combinations). Measured by raw token/storage footprint instead, the pair table requires only $\sim 2.5\times$ fewer tokens than storing $\mathcal{T}$ verbatim. We therefore do not claim a large quantitative compression, and instead base the “much weaker prior” argument on this qualitative distinction: $\mathcal{X} = \mathcal{R}$ carries none of the word order, syntax, or style information that makes $\mathcal{T}$ directly usable as training data. Under this weak prior, cross-domain performance can be further improved (Table 4), indicating that ToCa possesses good synthesis controllability and can effectively integrate additional lexical co-occurrence information when it happens to be available. We report ToCa_{S+ $\mathcal{X}$} purely as an oracle/weak-prior sensitivity study, not as a reduced-cost alternative to constructing $\mathcal{T}$: since $\mathcal{X} = \mathcal{R}$ is itself extracted from $\mathcal{T}$ ’s captions, this variant necessarily assumes some access to target-domain text, and it is not required for, nor part of, the main, strictly data-inaccessible cross-domain claim above, which relies solely on ToCa_S.

4.3. Data-efficient synthesis

In this section, we discuss text synthesis with low quantities of data. Following the ViECap setting, we randomly select data from different scales in the COCO training set for text synthesis and training. As shown in Table 5, regardless of the scale, ToCa significantly improves the performance of the model in both in-domain captioning and cross-domain captioning, with the maximum improvement being 20 CIDEr

Table 3
Flexible in-domain synthesis.

Method	(a) FlickrStyle10K								(b) MSR-VTT			
	Romantic				Humorous				Method	B@4	M	C
	B@1	B@3	M	C	B@1	B@3	M	C	MAGIC [10]	5.5	13.3	7.4
StyleNet [50]	13.1	1.5	4.5	7.2	13.4	0.9	4.3	11.3	CLIPRe [10]	10.2	18.8	19.9
MemCap [3]	21.2	4.8	8.4	22.4	19.9	4.3	7.4	19.4	DeCap [12]	23.1	23.6	34.8
CapDec [11]	21.4	5.0	9.6	26.9	24.9	6.0	10.2	34.1	CapDec [11]	22.2	22.5	29.4
ViECap [14]	25.7	6.5	10.4	33.6	24.3	6.5	10.4	35.0	ViECap [14]	23.3	21.5	29.3
+ToCa	27.6	8.5	11.7	41.4	26.7	8.0	11.5	41.5	+ToCa	28.8	24.2	36.5

Table 4

Transferable cross-domain synthesis. ToCa_S uses zero target-domain (COCO) information and is the primary cross-domain result; ToCa_{S+X} is a separate sensitivity study with an additional sparse lexical-pair prior (see Sections 3 and 4).

Method	Flickr30k $\Rightarrow$ COCO			
	B@4	M	C	S
MAGIC [10]	5.2	12.5	18.3	5.7
DeCap [12]	12.1	18.0	44.4	10.9
CapDec [11]	9.2	16.3	27.3	–
ViECap [14]	12.6	19.3	54.2	12.5
+ToCa _S (main, zero target info)	14.8	20.4	58.9	13.9
– sensitivity study with an additional weak lexical prior –				
+ToCa _{S+X}	15.5	21.3	61.5	14.8

Table 5

Practical data-efficient synthesis.

Data	Method	COCO	NoCaps val			
		test	In	Near	Out	Overall
0.01%	CapDec	0.5	0.1	0.1	0.1	0.1
	ViECap	0.3	0.1	0.1	0.1	0.1
	ToCa	23.4	17.3	18.9	18.0	19.4
0.1%	CapDec	24.0	13.2	11.0	6.2	10.4
	ViECap	32.3	20.9	27.6	34.9	30.2
	ToCa	49.0	35.1	39.3	39.7	40.6
1%	CapDec	55.8	29.6	20.5	9.8	18.9
	ViECap	63.9	34.6	39.9	39.3	40.4
	ToCa	80.3	53.7	58.2	59.0	59.9
10%	CapDec	83.6	47.3	39.8	19.1	35.4
	ViECap	83.4	45.9	51.8	48.7	53.3
	ToCa	91.8	59.2	62.5	60.4	63.0
100%	CapDec	92.7	60.1	50.2	28.7	45.9
	ViECap	92.9	61.1	64.3	65.0	66.2
	ToCa	95.0	64.6	69.1	70.5	70.9

score at the 1% scale. It is surprising that ViECap, based on ToCa's synthesized data augmentation, achieves comparable performance to ViECap with ten times the amount of data at any scale ($>0.1\%$). Even when the data volume is only 0.01% (**only 56 texts**), ToCa is still able to synthesize above 1k caption texts without repetition, greatly enhancing the captioning performance. These improvements are attributed to ToCa's structure and recombination strategy, which provide ample prompts for LLM synthesis of texts, even with small data volumes.

Stability across random subsets. Since the 0.01% and 0.1% scales involve only 56 and 566 captions, respectively, the particular random subset drawn from the COCO training set could itself have a substantial effect on the results. To verify that the reported improvements are not an artifact of a single favorable subset, we repeat the entire pipeline – subset sampling, template/lexical-pair extraction, LLM synthesis (for ToCa), and model training – across 3 independently drawn random subsets (different random seeds) at both the 0.01% and 0.1% scales for both ToCa and the baseline ViECap. Table 6 reports the mean $\pm$ (sample) standard deviation of CIDEr scores across these 3 repetitions.

Three main observations can be drawn. First, on the same 3 random subsets, ToCa consistently and substantially outperforms ViECap across

Table 6

Stability of data-efficient results. Both ToCa and ViECap are evaluated across 3 independent random subsets (mean $\pm$ std of CIDEr).

Data	Method	COCO	NoCaps Val			
		test	In	Near	Out	Overall
0.01%	ViECap	0.3 $\pm$ 0.1	0.1 $\pm$ 0.0	0.1 $\pm$ 0.0	0.1 $\pm$ 0.0	0.1 $\pm$ 0.0
	ToCa	19.9 $\pm$ 3.0	13.7 $\pm$ 3.1	13.5 $\pm$ 4.7	11.5 $\pm$ 5.6	13.5 $\pm$ 5.1
0.1%	ViECap	31.8 $\pm$ 1.2	20.5 $\pm$ 0.8	27.1 $\pm$ 1.1	34.2 $\pm$ 1.5	29.7 $\pm$ 1.2
	ToCa	49.5 $\pm$ 4.0	33.3 $\pm$ 0.6	39.5 $\pm$ 1.7	39.2 $\pm$ 3.1	40.5 $\pm$ 2.1

all metrics and scales. At the 0.01% scale, ViECap consistently fails to form a functional captioning model (CIDEr ~ 0.1 – 0.3 across seeds), whereas ToCa achieves an average CIDEr score of 19.9 on COCO and 13.5 overall on NoCaps Val, confirming that ToCa's dramatic performance jump in extreme low-data regimes is robust to subset variation. Second, the variance for ToCa is noticeably smaller at the 0.1% scale (std 0.6–3.1 across metrics) than at 0.01% (std 3.0–5.6), consistent with the intuition that a larger seed corpus yields a more stable lexical-pair and structure-template distribution for synthesis. Third, ViECap shows relatively low variance at 0.1% (std 0.8–1.5), but its performance ceiling remains substantially lower than ToCa (e.g., COCO CIDEr 31.8 ± 1.2 vs. 49.5 ± 4.0), proving that ToCa's superiority stems from its structural synthesis strategy rather than a fortuitous subset selection.

Three observations follow. First, even the worst of the 3 seeds at 0.01% (COCO CIDEr 18.0) remains an order of magnitude above the ViECap baseline (0.3), so the qualitative conclusion – ToCa dramatically improves data-efficient captioning – is robust to subset choice, even though the single-run value reported in Table 5 (23.4) is on the higher end of the observed range and the mean (19.9) is a more representative estimate. Second, the variance is noticeably smaller at the 0.1% scale (std 0.6–3.1 across metrics) than at 0.01% (std 3.0–5.6), consistent with the intuition that a larger seed corpus yields a more stable lexical-pair/structure-template distribution for synthesis. Third, this pattern holds consistently across all five metrics (COCO, In, Near, Out, Overall), not just the two we highlighted earlier, indicating that the improvement and its variance are not an artifact of any single evaluation split.

4.4. Ablation studies

The effect of synthesis volume. We conduct experiments to evaluate the performance of model under different synthetic data volumes, as shown in Fig. 3. The observation showed a steady improvement in model performance from 0.1M to 3M data under $D + T$, but it reached a plateau beyond 3M. We believe that ToCa is still capable of generating diverse novel texts (based on Eq. (13)), but not relevant to the target dataset, which leads to no performance gain. The experimental results also support this statement: in Fig. 3(a), models trained only on D show a decreasing CIDEr score on COCO as the data volume increases. This is because after synthesizing strongly correlated and high-frequency texts, further increasing the quantity leads to weakly correlated and low-frequency novel texts (noise). In contrast, Fig. 3(b) shows more stability on NoCaps, as NoCaps is not part of the corpus data, so the correlation between the synthesized text and NoCaps captions remains

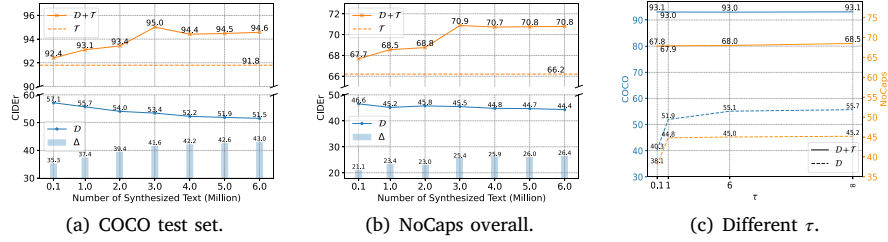


Fig. 3. The effect of the number of synthesized text and different τ . $D+T$ represents training on synthesized data D and fine-tuning on accessible target data T . D represents training solely on synthesized data. Δ denotes the difference in CIDEr scores between $D+T$ and D .

Table 7

The effect of lexical pairs $\mathcal{R}$ and structure templates $\mathcal{G}$. ICS indicates in-context synthesis that generates texts with reference examples.

Method	Data	COCO				
		test	NoCaps val			
			In	Near	Out	Overall
Baseline	T	91.8	60.9	63.7	64.2	65.4
ICS	$D+T$	92.7	60.0	65.1	67.2	67.1
w/o $\mathcal{R}$	D	19.9	13.0	16.1	14.6	15.9
	$D+T$	92.2	61.0	64.7	65.2	66.5
w/o $\mathcal{G}$	D	54.3	39.3	42.7	41.7	43.2
	$D+T$	93.0	61.5	66.4	67.2	68.3
ToCa	D	55.7	42.7	44.1	43.7	45.2
	$D+T$	93.1	61.8	66.5	68.4	68.5

stable regardless of the scale of synthesis. The higher CIDEr score on COCO compared to NoCaps also indicates that ToCa does indeed approach the target dataset to some extent.

The effect of deconstruction strategy. In order to investigate the impact of deconstructing captions into lexical pairs $\mathcal{R}$ and structure templates $\mathcal{G}$ on synthesized text, we conducted experiments on a dataset of 1M, as shown in Table 7. It can be observed that the removal of $\mathcal{R}$ results in a significant decrease in performance on the dataset D , indicating the crucial importance of lexical pairs as the core meaning of captions for synthesizing text in the target domain. Although the removal of $\mathcal{G}$ does not lead to a significant decrease on the dataset $D+T$, there is a decline on the dataset D , suggesting that structure templates also play a controlling role in synthesizing text in the target domain. Comparing the two ablations directly, removing $\mathcal{R}$ causes a much larger degradation on D (CIDEr 55.7 $\rightarrow$ 19.9) than removing $\mathcal{G}$ (CIDEr 55.7 $\rightarrow$ 54.3). This confirms that lexical pairs, which encode entity co-occurrence and thus the core semantic content of a caption, are the primary driver of synthesis quality, whereas structure templates mainly regulate syntactic diversity and fluency rather than semantic correctness. In practice, when the curation or computational budget is limited, priority should be given to constructing a reliable lexical-pair table over a diverse structure-template set. By considering Eq. (12) in conjunction with Fig. 3(c), it becomes apparent that selecting a smaller value for τ can result in an excessive diversification of the synthesized data, thereby straying too far from the realm of τ and introducing noise that ultimately diminishes performance. Although training on $D+T$ mitigates a substantial portion of these discrepancies, $\tau = \infty$ is chosen by default for more robust performance, as it aligns more closely with our objective function Eq. (1).

4.5. Quality control of synthesized text

As described in Section 3, ToCa filters out any LLM completion s'' that fails to contain all required lexical tokens $\{w_0, \dots, w_n\}$. However, satisfying this lexical constraint does not guarantee that a sentence is fluent or semantically plausible. We therefore report two additional quality-control analyses.

Table 8

Filtering rate of the lexical-containment filter, at the two data-efficient scales.

Scenario	Filtering rate
Data-efficient (0.01%, 56 seed captions)	45.3%
Data-efficient (0.1%, 566 seed captions)	0.8%

Table 9

5 randomly sampled synthesized sentences (post-filtering) at each data-efficient scale, for qualitative inspection.

Scale	Randomly sampled sentence
0.01%	The living room is adorned with wooden furniture on its polished floor. A man watching the snowfall outside a window as bears hibernate in their dens.
	A woman sharing pizza and laughing with her friends against the backdrop of an outdoor setting. Several people are smiling at each other with smiles of genuine warmth and appreciation. The deck and story of an old bus are adorned with intricate carvings in the jungle setting.
	A man riding a horse over a bridge on a sunny morning. Several people dancing in a nightclub.
	The corner of Elm Street bears a sign with a brown horse and the words, 'General Store'. A soccer player kicks a ball during a match in the middle of the field. A white plate is placed on the table with a vase of flowers behind it.

Filtering rate. Table 8 reports the proportion of LLM completions discarded by the lexical-containment filter. We focus this analysis on the data-efficient scenario, as it spans the two extremes of seed-corpus size (56 vs. 566 captions) studied in this paper, and is the setting where filtering behavior is most sensitive to data scarcity.

We observe that the filtering rate is markedly higher at the smallest data-efficient scale (0.01%, 45.3%) than at the 0.1% scale (0.8%). This is consistent with the combinatorial nature of ToCa's sampling process (Eq. (12)): with only 56 seed captions, the lexical-pair and structure-template pools are much sparser, so a larger fraction of the entity combinations that get sampled together are difficult for the LLM to weave into a single fluent sentence within the retry budget, whereas the substantially larger 566-caption pool at 0.1% offers far more compatible template/lexical-pair combinations and is filtered far less aggressively. Together, these two scales bracket the filtering behavior observed across our experiments: the failure rate is bounded above by the most data-scarce setting we study.

Qualitative examples. Note that containing the required lexical tokens does not by itself guarantee that a sentence is fluent or plausible. Rather than relying solely on curated examples, we report 5 randomly sampled synthesized sentences at each of the two data-efficient scales in Table 9, without any manual selection or filtering beyond the lexical-containment step already described. Readers can directly judge typical output quality from these unfiltered, non-curated samples.

Most of the displayed samples are fluent and semantically coherent, while the examples also reveal occasional atypical entity combinations,

Table 10
Statistical measures between D and $\mathcal{T}$.

	ToCa on COCO					Flickr30k on COCO				
	P	R	P_w	R_w	Cosine	P	R	P_w	R_w	Cosine
Token	43.8	59.0	99.4	99.6	92.7	73.9	23.5	98.8	47.2	10.6
Structure	3.9	48.1	66.3	93.5	94.2	26.7	19.0	81.6	87.3	91.3

especially at the smaller 0.01% scale: e.g., “A man watching the snow-fall outside a window as bears hibernate in their dens” juxtaposes two only loosely related COCO entities (a person-at-a-window scene and a hibernating-bear scene), and “The deck and story of an old bus are adorned with intricate carvings in the jungle setting” combines a vehicle-related structure template with an unrelated “jungle” setting. This is consistent with the higher filtering rate observed at this scale (Table 8) and with our token-diversity analysis (Eq. (12), Section 4 ablation): sparser lexical-pair statistics at small scales occasionally lead the sampler to pair less compatible entities, and the LLM sometimes still manages to force them into one fluent sentence despite passing the lexical filter. Occasional atypical combinations of this kind are an expected trade-off of maximizing lexical diversity (controlled by τ in Eq. (12)), and are quantitatively analyzed in the ablation above.

4.6. Qualitative visualization

We visually showcase the qualitative results of the synthesized text and captioning results, as depicted in Fig. 1. On the left side of Fig. 1, the synthesizing results demonstrate the ability of the generated text to effectively incorporate entity information, adapt to different styles and tasks. On the right side of Fig. 1, the captioning results reveal the improved descriptive outcomes achieved through the process of enhanced training. These findings intuitively underscore the potential and advantages of text synthesis in elevating the quality and relevance of caption generation.

4.7. Distance between D and $\mathcal{T}$

To visually showcase the distance between the synthesized data by ToCa and the distribution of the target data, t-SNE [52] is employed to visualize the text features encoded by CLIP in D and $\mathcal{T}$ as shown in Fig. 4. Taking into consideration computational efficiency, a random selection of 5667 text features from COCO is utilized as the analytical sample. The distribution relations between the ToCa-synthesized data and COCO data are visualized across a relative quantity range from 0.1 $\times$ to 1 $\times$ to 10 $\times$, as depicted in Fig. 4(d)–(f). For ease of observation, the distributional relationship of data features between the two datasets, Flickr30k and COCO, used in the cross-domain captioning setting, is also visualized and presented as a reference. Upon comparing Fig. 4(a)–(c) and (d)–(f), it becomes evident that the degree of overlap between the synthesized data of ToCa and the distribution of COCO data is significantly higher than the overlap between the distribution of Flickr30k and COCO data. Simultaneously, upon observing the transition from (d) to (f), with an increase in data volume, ToCa exhibits an increasing number of novel samples. However, the proximity of these novel samples to COCO surpasses that of Flickr30k and COCO. Consequently, this outcome can qualitatively affirm that ToCa represents an effective solution for solving the objective Eq. (1), wherein it not only approximates the target data but also generates novel samples surpassing the target data.

In addition to the qualitative analysis of text feature visualization and the quantitative results regarding the final captioning performance, we have also conducted statistical measurements, as shown in Table 10, as a direct evaluation of the quantitative indicators for assessing the distance between the generated data and the target dataset. To facilitate analysis, we have also included the statistical measurements of

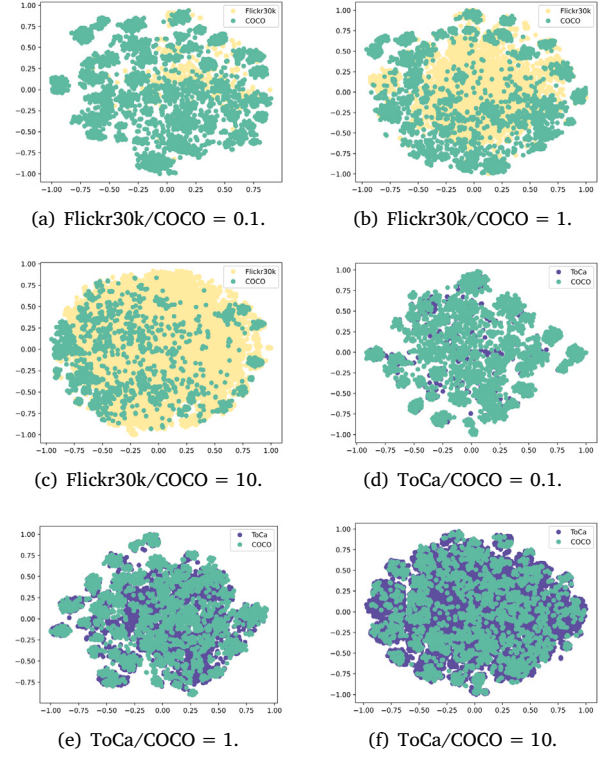


Fig. 4. t-SNE visualizations of Flickr30k, COCO, and ToCa. (a)–(c) and (d)–(f) respectively represent the variations in the distribution relations of features between Flickr30k and COCO, as well as between ToCa and COCO, across a relative quantity range from 0.1 $\times$ to 1 $\times$ to 10 $\times$.

Flickr30k on COCO as a reference for comparison. The statistical measurement metrics include precision (P) and recall (R) for both tokens and structures. Additionally, weighted precision (P_w) and weighted recall (R_w) are computed, taking into account the frequency. Furthermore, the cosine similarity (Cosine) is calculated for the frequency distribution of tokens and structures between the two datasets. From Table 10, it can be observed that the (P_w) and (R_w) metrics, which take frequency into consideration, are significantly higher than the P and R metrics. This indicates that COCO dataset consists of both high-frequency and low-frequency data, and ToCa effectively adheres to these characteristics. Comparing the results of ToCa and Flickr30k, it can be observed that the level of approximation between the synthesized data by ToCa and COCO is higher than that of Flickr30k and COCO (as indicated by higher R_w and Cosine values for ToCa). Additionally, it is evident that the token-based Cosine for Flickr30k and COCO involve descriptions of different domains of images, resulting in significant token differences. The higher structure-based Cosine can be attributed to the fact that both datasets involve captioning tasks, where the text follows a caption-style format. Both token and structure metrics for the data generated by ToCa exhibit high values (R_w and Cosine). Furthermore, it can be observed that the P values for both tokens and structures are significantly lower for ToCa compared to Flickr30k. This is primarily due to the larger volume of data generated by ToCa, resulting in the generation of more novel tokens and structures.

5. Conclusion

In this paper, we propose ToCa, which explores the use of an LLM to synthesize caption text under a purely text-only setting. This approach

has yielded conspicuous performance gains and cost reductions across various real-world challenges and a diverse range of caption tasks. We perceive LLM as a comprehensive repository, encapsulating a multitude of knowledge, while our algorithm constructs a refined cue to retrieve suitable information. Consequently, our goal for ToCa goes beyond enhancing visual captioning. We aim to inspire diverse text generation tasks and data synthesis from text prompts. In the future, we will explore and refine ToCa's application in these domains, promoting the achievement of the paradigm.

Limitations. ToCa is positioned as a practical solution for settings with constrained data accessibility, i.e., limited or no target-domain annotations and limited compute for data creation, rather than as a method intended to surpass every existing approach in absolute performance under unconstrained conditions. When full paired image–text annotation or large-scale paired image–text synthesis is affordable, dedicated in-domain supervised methods or larger-scale paired-synthesis pipelines may still achieve higher absolute captioning performance. Furthermore, since text synthesis relies on an LLM to complete masked templates, the quality of D is ultimately bounded by the completion ability of the chosen LLM, and, as discussed in Section 4.5, a small fraction of generated sentences may still exhibit atypical or implausible entity combinations, particularly at low τ .

CRedit authorship contribution statement

Qing Zhou: Writing – original draft, Visualization, Validation, Formal analysis. **Junlin Huang:** Visualization, Validation, Methodology. **Junyu Gao:** Writing – review & editing, Resources. **Qi Wang:** Supervision, Resources.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT in order to enhance the readability of the article. After using this tool, the author(s) reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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